PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	A practical and validated tool to assess falls risk in the primary care setting: A systematic review
AUTHORS	Meekes, Wytske; Korevaar, J; Leemrijse, Chantal; Van de Goor, Ien

VERSION 1 – REVIEW

REVIEWER	Renfro, Mindy University of Montana System
REVIEW RETURNED	25-Nov-2020

GENERAL COMMENTS	Fall risk screening tools and fall risk tests are different and should be identified and defined separately from each other. You have spoken of screening instruments but reported on fall risk test instruments. The fall risk tests you chose were developed for use primarily as clinical outcome measures for physical therapists (aka physiotherapists) and not for primary care practitioners and/or their staffs. However, the CDC's STEADI toolkit (validation attached) was developed to exactly meet the need of the primary care provider and can be used successfully by the PCP's staff, public health, nurses, etc. The STEADI is based upon the work of Laurence Z. Rubenstein and David A Ganz - both PCPs and was vetted by a team of 50 expert practitioners in 2009 including MD, PT, OT, RN, PharmD, etc. I am curious why this validated tool was not considered if the systematic review truly covers all of the available literature from the stated time period. Please see https://images.app.goo.gl/iM7fipVTQpBKYR1o8 and https://pubmed.ncbi.nlm.nih.gov/22124417/ and https://pubmed.ncbi.nlm.nih.gov/28947669/ From what I can tell, the STEADI meets your inclusion criteria. I also don't see Lusardi and Fritz's recent SR on the same topic (attached) which is here https://pubmed.ncbi.nlm.nih.gov/27537070/#&gid=article-figures&pid=figure-2-uid-1 and draws different conclusions from yours. Please consider the contrast in your discussion and limitations of your study.

REVIEWER	Ambrens, Meghan CQUniversity, School of Health, Medical and Applied Sciences
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	Abstract No registration details (PROSPERO) provided.

	There is very limited information under Design (line 22). The methods do not provide an indication of the inclusion criteria, or search limitations. There is a spelling error with the word Cochrane (line 23) Entire manuscript: Please spell out all acronyms with first use e.g. line 69 - AGS/BGS/AAOS. Use of capital letters for fall history. Introduction: Provides a clear aim and explains the problem or background sufficiently. More information is needed to explain the current knowledge related to the research question. Missing is comment on a previous Systematic Review by Gates et al (2008). What new information does this Review add to the body of knowledge? Also, missing is the recognition that both the British Geriatric Society and the American Geriatric Society recommend that GPs should screen their patients for falls risk each year. Line 56 - the consequences of falls needs a reference. Methods: Please provide a justification for the timeframe included in the search strategy - especially given the 2008 Systematic Review by Gates et al. A robust way to conduct a study selection usually requires two blinded reviewers. While it is acceptable to use one person during the initial screening of titles and abstracts the Cochrane Handbook for Systematic Reviews of Interventions recommends as ideal the use of two at least two people working independently through each stage of screening. Analysis: Please provide a reference to support the criteria regarding suitability of a fall risk screening instrument for the primary care setting. I assume this has been developed based on the feedback from focus groups. Line 112: We don't really 'develop a fall'. Please change wording to reflect a fall event to 'experience a fall' or 'having a fall'. Line 123: please change taken to took Line 125: Please delete the word 'also' Line 125: please change suitable to suitability Lines 126 and 129: please change taking into taken Patient and Public Involvement The authors are to be congratulated on consulting primary care professionals as end users. Could you be more explicit regarding the outcome of this consultation, specifically how it was used in the design or analysis of this Review. Results: Please include a paragraph about the participants included in the studies, including gender balance and country of study. Please report consistently the number of studies that included each scale Please include the original citations of each of the scales/tests you have reported in the results.

	Line 145 consider wording change to time taken not time measured Discussion: The reason people fall is multi-factorial. An exploration of other factors in this discussion is warranted. This Review has included tests/screens that focus on balance and mobility as the predictor of falls, therefore it is difficult to predict falls if using a screening tool that only focusses on balance and mobility. This means that other risk factors for falls (e.g. cataracts or medications) have not been considered, and therefore people with these risk factors are not being screened. Furthermore, a review of the Systematic Review by Barry et al (2014) should be taken into consideration. Barry et al (2014). Is the Timed Up and Go test a useful predictor of risk of falls in community dwelling older adults: a systematic review and meta-analysis. BMC Geriatrics 14:4. http://www.biomedcentral.com/1471-2318/14/14 Line 206: please change fast to vast Line 207: please report the mean intervention period in addition to the range Lines 242-243: please provide a reference Line 244: please provide a reference to support this claim Lines 249-252: please provide literature to support this? Practice recommendations: Please clarify what you mean by the association between sitting and fall risk. Table 1: All tables should be standalone. Please provide a complete and thorough key/legend for all units of measure, abbreviations, acronyms The POMA gait and balance measures should include their acronym for consistency The Tiedemann study has a spelling mistake - please change to ≥fall

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Mindy Renfro, University of Montana System Comments to the Author:
*** This reviewer has provided some supplements to their review. Please find them in the attached files

- Fall risk screening tools and fall risk tests are different and should be identified and defined separately from each other. You have spoken of screening instruments but reported on fall risk test instruments.

Thank you for pointing this out. We changed the text to make clear these are prognostic tests and not screening tools.

- The fall risk tests you chose were developed for use primarily as clinical outcome measures for physical therapists (aka physiotherapists) and not for primary care practitioners and/or their staffs.

Indeed, many included assessment tools were initially created to be used by physical therapists, however they can also be used by other primary care providers with the appropriate training.

• However, the CDC's STEADI toolkit (validation attached) was developed to exactly meet the need of the primary care provider and can be used successfully by the PCP's staff, public health, nurses, etc. The STEADI is based upon the work of Laurence Z. Rubenstein and David A Ganz - both PCPs and was vetted by a team of 50 expert practitioners in 2009 including MD, PT, OT, RN, PharmD, etc. I am curious why this validated tool was not considered if the systematic review truly covers all of the available literature from the stated time period. Please see
https://images.app.goo.gl/iM7fipVTQpBKYR1o8 and
https://pubmed.ncbi.nlm.nih.gov/22124417/ and
https://pubmed.ncbi.nlm.nih.gov/28947669/
From what I can tell, the STEADI meets your inclusion criteria.

Thank you for your suggestion. The first link appears to be a volume/book which includes the following articles:
- Cardiac causes for falls and their treatment / Hilary Cronin and Rose Anne Kenny
- Vision and falls in older people : risk factors and intervention strategies / Stephen R. Lord, Stuart T. Smith, and Jasmine C. Menant
- Medication and falls : risk and optimization / Nichola Boyle, Vasi Naganathan, and Robert G. Cumming
- The role of exercise in fall prevention for older adults / Debra J. Rose and Danielle Hernandez
- Environmental assessment and modification as fall-prevention strategies for older adults / Jon Pynoos, Bernard A. Steinman, and Anna Q.D. Nguyen
- Preventing falls and fall-related injuries in hospitals / David Oliver, Frances Healey, and Terry P. Haines
- Public health approach to fall prevention among older persons in Canada / Vicky Scott, Brandon Wagar, Alison Sum, Sarah Mefcalfe, and Lori Wagar
- Comprehensive approach to fall prevention on a national level : New Zealand / A. John Campbell and M. Clare Robertson
- Fall prevention in Australia : policies and activities / Lindy Clemson, Caroline F. Finch, Keith D. Hill, and Gill Lewin
- A cost-effectiveness analysis of a proposed nation falls prevention program / Shinyi Wu, Emmett B. Keeler, Laurence Z. Rubenstein, Margaret A. Maglione, and Paul G. Shekelle
- An older adult falls research agenda from a public health perspective / Judy A. Stevens, Grant T. Baldwin, Michael F. Ballesteros, Rita K. Noonan, and David A. Sleet.

It is unclear for us which article is about the STEADI when reading the titles and abstracts.
The second link, “Multifactorial screening for fall risk in community-dwelling older adults in the primary care office: development of the fall risk assessment & screening tool” of Renfo, appears not to be about the STEADI but about the FRATS. Still, the study of Renfo would have been excluded as the sensitivity/specificity of the tool were not prospectively investigated.
The third link, “Operationalization and Validation of the Stopping Elderly Accidents, Deaths, and Injuries (STEADI) Fall Risk Algorithm in a Nationally Representative Sample” of Lohman et al., was initially eligible to include however, we did not find in our search 3 articles concerning the STEADI that fulfilled all our eligibility criteria. Therefore, the STEADI in the end was not eligible to include in this review.

Exclusion of tools does not mean that these tools are not suitable or are not validated. There are indeed many other tools assessment tools available to assess fall risk, but they did not meet our eligibility criteria:

Articles were included when they met the following inclusion criteria:
1. Prospective studies in which the primary or secondary purpose was to evaluate the performance of one or more fall risk assessment tools for predicting fallers.
2. The participants were older people living in the community or substantially independently
3. Full articles published in English, Dutch or German

Articles were excluded when they met one or more of the following exclusion criteria:
1. Fall risk assessment tools which require expensive computer software programs, other advanced expensive software or instruments not available in usual primary care units (e.g. sensors, electronical platforms, force plates).
2. Literature reviews and studies with no follow up of fall incidents.
3. No reported Area Under the Curve (AUC), sensitivity or specificity of the fall risk assessment tools.
4. Assessment tools specifically developed for or only tested on populations with a specific disease (e.g. cancer, diabetes, Parkinson etc.)
5. The participants were living in hospital or other institutionalised settings
Additionally, only the tools that have been assessed in at least three different studies were included in the final analysis to ensure validity of the included tool as studies may differ, for example in selected population in age, sex, or frailty.)

Note. Even when the STEADI was included, results would not have changed as the AUC for the STEADI (AUC=0.641) was insufficient when taking the threshold of Šimundić for good diagnostic accuracy into account (AUC>0.7) (based on the study of Lohman et al.).

- I also don't see Lusardi and Fritz's recent (2017) SR on the same topic (attached) which is here https://pubmed.ncbi.nlm.nih.gov/27537070/#&gid=article-figures&pid=figure-2-uid-1 and draws different conclusions from yours. Please consider the contrast in your discussion and limitations of your study.

Thank you for suggesting this interesting article. We agree with Lusardi and Fritz's statement, "falls in later life are multi-factorial, identifying those older individuals living in the community who are most likely to fall is problematic." According to the Lusardi and Fritz, "No single test/measure demonstrated strong PoTP values". Also in our study, we did not find a (single) assessment tool with good predictive performance.
We believe that Lusardi and Fritz have good recommendations based on their study results however they seem not to consider another important factor into account when selecting fall risk assessment tools, namely the suitability of the assessment tools for implementation in the care providers' daily routine.
We believe the focus of Lusardi and Fritz' article is different from our study as they did not take the suitability for implementation into account and because they look at the posttest probability of tools. Therefore, we believe this article does not fit our Discussion section.

Reviewer: 2
Miss Meghan Ambrens, CQUniversity
Comments to the Author:

Abstract

- No registration details (PROSPERO) provided.

Thank you for this suggestion. We did not register this review before conducting data extraction. Unfortunately, PROSPERO only accepts reviews provided that data extraction has not yet started (since 1st October 2019). Therefore, we are now unable to register. We will take your advice into account for future reviews.

- There is very limited information under Design (line 22).

We added information to Design, see Abstract, Design (considering the word limit of 300 words).

- The methods do not provide an indication of the inclusion criteria, or search limitations.

We added eligibility criteria to the text, see Abstract, Design and Methods.

- There is a spelling error with the word Cochrane (line 23)

We corrected the misspelled word.

Entire manuscript:

- Please spell out all acronyms with first use e.g. line 69 - AGS/BGS/AAOS.

We checked and revised the text, see Introduction.

- Use of capital letters for fall history.

We refer in the text to the fall history assessment tool as 'Fall History (FH)' and to fall history in general as 'fall history'. We checked and revised the text.

Introduction:

- Provides a clear aim and explains the problem or background sufficiently.

Thank you.

- More information is needed to explain the current knowledge related to the research question. Missing is comment on a previous Systematic Review by Gates et al (2008). What new information does this Review add to the body of knowledge? Also, missing is the recognition that both the British Geriatric Society and the American Geriatric Society recommend that GPs should screen their patients for falls risk each year.

We revised the text by adding information about the review of Gates et al., see Introduction, paragraph 3. We have added a sentence regarding annually assessing fall risk to the Discussion section, Practice recommendations.

- Line 56 - the consequences of falls needs a reference.

We added two references (4,5), see Introduction.

Methods:

- Please provide a justification for the timeframe included in the search strategy - especially given the 2008 Systematic Review by Gates et al.

The proportion of older people is increasing and the current population of older people age differently compared to 20 years ago (e.g. people get older, different comorbidities, more active). So, it is

important that suitable fall risk assessment tools are validated in the current population of older people. Therefore, we believe the range 2000-2020 is best suitable for this review.
We adjusted the text, see Methods, paragraph ‘Eligibility criteria and study selection’.

- A robust way to conduct a study selection usually requires two blinded reviewers. While it is acceptable to use one person during the initial screening of titles and abstracts the Cochrane Handbook for Systematic Reviews of Interventions recommends as ideal the use of two at least two people working independently through each stage of screening.

We agree that this would be the ideal as recommended by the Cochrane Handbook. However, this was not feasible within this project. Therefore, we described the study selection in complete detail.

Analysis:

- Please provide a reference to support the criteria regarding suitability of a fall risk screening instrument for the primary care setting. I assume this has been developed based on the feedback from focus groups.

These criteria were indeed based on an informal focus group discussion (see patient and public involvement) but also based on previous literature. We added 4 references to the remaining criteria.

- Line 112: We don't really 'develop a fall'. Please change wording to reflect a fall event to 'experience a fall' or 'having a fall'.

We changed ‘develop’ to ‘experiencing’.

- Line 123: please change taken to took

We changed ‘taken’ to ‘taking’.

- Line 125: Please delete the word 'also'

We deleted the word ‘also’.

- Line 125: please change suitable to suitability Lines 126 and 129: please change taking into taken

We changed ‘suitable’ to ‘suitability’.

Patient and Public Involvement

- The authors are to be congratulated on consulting primary care professionals as end users. Could you be more explicit regarding the outcome of this consultation, specifically how it was used in the design or analysis of this Review.

We adjusted the text, see paragraph ‘Patient and public involvement’.
The results from this informal focus group discussion (participants: 4 GPs, 2 practice nurses, 3 district nurses), together with previous literature, defined the suitability criteria used in this study.
(Later on, also a small pilot with our selected fall risk assessment tool was conducted at two GP practices and subsequently an implementation study was conducted among 12 GP practices.)

Results:

- Please include a paragraph about the participants included in the studies, including gender balance and country of study.

This overview can be found in Additional file 2.

- Please report consistently the number of studies that included each scale

We changed ‘13’ to ‘14’ articles for the TUG test.

- Please include the original citations of each of the scales/tests you have reported in the results.

When adding these 7 references, we believe the reference list will become too long. In addition, the current references already describe the assessment tools in detail.
If the editor also prefers adding the original references, we will include them.

Line 145 consider wording change to time taken not time measured

We changed ‘measured’ to ‘taken’.

Discussion:

- The reason people fall is multi-factorial. An exploration of other factors in this discussion is warranted. This Review has included tests/screens that focus on balance and mobility as the predictor of falls, therefore it is difficult to predict falls if using a screening tool that only focusses on balance and mobility. This means that other risk factors for falls (e.g. cataracts or medications) have not been considered, and therefore people with these risk factors are not being screened.

Thank you for pointing this out. Indeed, assessing high fall risk by using fall history does not directly include other fall risk factors. However, having a fall history indicates that there are risk factors/underlying causes which require treatment (medication review, balance/strength training, nutrition/different diet, referral to optician etc.).
We revised the text, see Discussion, Practice Recommendations and Further Research.

- Furthermore, a review of the Systematic Review by Barry et al (2014) should be taken into consideration.

Barry et al (2014). Is the Timed Up and Go test a useful predictor of risk of falls in community dwelling older adults: a systematic review and meta-analysis. BMC Geriatrics 14:4. http://www.biomedcentral.com/1471-2318/14/14
We did refer to Barry, see: “The results of this review are also acknowledged by others. For example, even though the TUG test is widely used to assess fall risk, other studies also showed the lack of predictive ability of the TUG test regarding falls [50, 51]” (reference 50 = Barry et al, 2014)

- Line 206: please change fast to vast

We changed ‘fast’ to ‘vast’.

- Line 207: please report the mean intervention period in addition to the range

We added the mean intervention period “(mean 15 months; range 6 month-9 years)”

- Lines 242-243: please provide a reference

We added two references (68,69).

- Line 244: please provide a reference to support this claim

We added a reference (70).

- Lines 249-252: please provide literature to support this?

We added 6 references (15-18 and 73,74).

Practice recommendations:

- Please clarify what you mean by the association between sitting and fall risk.

We revised the text: “Even if the patient says ‘no’, the GP might still notice a high fall risk based on clinical judgement (e.g. walking or sitting difficulties due to strength and balance problems, dizziness, use of benzodiazepines, visual impairment, etc.).”, see Discussion section, Practice recommendations.

Table 1:

- All tables should be standalone. Please provide a complete and thorough key/legend for all units of measure, abbreviations, acronyms The POMA gait and balance measures should include their acronym for consistency

We removed the abbreviations from Table 1 and added thorough legends to both tables.

- The Tiedemann study has a spelling mistake - please change to ≥fall

We changed ‘fal’ to ‘fall’.

VERSION 2 – REVIEW

REVIEWER	Ambrens, Meghan CQUniversity, School of Health, Medical and Applied Sciences
REVIEW RETURNED	30-Mar-2021

GENERAL COMMENTS	I acknowledge the work the authors have put into the manuscript. This manuscript would benefit from having a native English speaker review it - throughout the manuscript there are numerous instances where word choice is either incorrect or has been incorrectly used, there errors with the singular and plural use of tool/tools, grammatical errors, and sentence structure errors i.e., starting a sentence with ‘Because’. Quotations have not followed the correct convention for quotations using the BMJ referencing style. The BMJ style follows the Vancouver referencing system for quotations. i.e., remove italics, provide a page number. This paper has a number of significant methodological limitations that need to be discussed in the limitations. Firstly, it was not registered on PROSPERO. Secondly, study selection was not undertaken as a blinded review. The Cochrane Handbook for Systematic Reviews of Interventions recommends as ideal the use of two at least two people working independently through each stage of screening. Falls risk assessment is difficult and perhaps the findings from this paper highlight this complexity? I acknowledge that the authors conclude a falls history is perhaps the most suitable tool for the primary care setting. A falls history is a screening tool rather than an assessment tool. The difference between these types of tools should be addressed in the discussion and the (clinical) implications considered as part of future research.

	I am finding it difficult to clearly understand what new evidence this paper adds since the Gates et al (2008) review. I think the authors need to consider putting their discussion into context in terms of what the implications of their results are for clinicians and future research. Perhaps they would benefit from answering the question 'if a clinician reads your paper, what is the message they take away from it'. Making this clear would strengthen this manuscript.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Miss Meghan Ambrens, CQUniversity

Comments to the Author:

I acknowledge the work the authors have put into the manuscript.

-This manuscript would benefit from having a native English speaker review it - throughout the manuscript there are numerous instances where word choice is either incorrect or has been incorrectly used, there errors with the singular and plural use of tool/tools, grammatical errors, and sentence structure errors i.e., starting a sentence with 'Because'.

 We sent the manuscript for editing to a native speaker from a language center. We made corrections throughout the text.

-Quotations have not followed the correct convention for quotations using the BMJ referencing style. The BMJ style follows the Vancouver referencing system for quotations. i.e., remove italics, provide a page number.

 We revised the style for quotations following the Vancouver style. For example:

…'since it has no clinical consequences other than the waste of scarce nursing resources' [72, p421].

-This paper has a number of significant methodological limitations that need to be discussed in the limitations. Firstly, it was not registered on PROSPERO. Secondly, study selection was not undertaken as a blinded review. The Cochrane Handbook for Systematic Reviews of Interventions recommends as ideal the use of at least two people working independently through each stage of screening.

 Thank you for pointing this out. We revised the text and added these methodological limitations to the section Discussion, Strengths and Limitations:

“This review was not registered at PROSPERO, the international prospective register of systematic reviews. This could have caused duplication of review topics. Nonetheless, no ongoing reviews were found in the PROSPERO register that specifically focus on suitability of falls risk assessment tools for the primary care setting.

In this review, the initial screening of titles and abstracts was performed by one researcher (WM). For the second round of selection, a sample of 200 articles was reviewed independently by a second researcher (JK, CL or IG), based on abstract (>95% consensus). Even though this is an acceptable procedure according to the Cochrane Handbook for Systematic Reviews of Interventions, each screening step should ideally be performed by at least two people working independently [80]. Our results might therefore be subject to bias due to our method of study selection.”

-Falls risk assessment is difficult and perhaps the findings from this paper highlight this complexity? I acknowledge that the authors conclude a falls history is perhaps the most suitable tool for the primary care setting. A falls history is a screening tool rather than an assessment tool. The difference between these types of tools should be addressed in the discussion and the (clinical) implications considered as part of future research.

 We agree that it can be confusing talking about screening or assessment tools and high or low risk. We therefore added our definition of assessment tool to the section Introduction and revised the text accordingly in the section Discussion.

“It should be noted that, in this study, FH is defined as an assessment tool and not as a screening tool. A falls risk assessment tool defines the nature of the problem, and thus whether a patient is or is not at high risk of falling [21]. No additional assessment is required to identify high or low falls risk. Additional assessment (e.g. multifactorial assessment) is needed only to determine which intervention is needed in order to reduce a patient’s high falls risk. Screening tools are intended to evaluate the possible presence of specific problems. A screening tool would require additional assessment in order to verify that a patient has a high falls risk [21].”

-I am finding it difficult to clearly understand what new evidence this paper adds since the Gates et al (2008) review. I think the authors need to consider putting their discussion into context in terms of what the implications of their results are for clinicians and future research. Perhaps they would benefit from answering the question ‘if a clinician reads your paper, what is the message they take away from it’. Making this clear would strengthen this manuscript.

 Compared to Gates et al. 2008, our study aims to find a tool suitable for primary care providers to identify patients with a high fall risk. Our message is partly the same as Gates, there is no conclusive evidence that one tool is the best to identify high or low fall risk. However, the tool that meets relevant criteria for primary care professionals (limited time and lack resources for expensive equipment, room or training) is patient’s fall history, which can be best integrated in their daily practice.

To highlight what our review adds to the review of Gates et al. (2008) we added to the section Discussion: “It is therefore impossible to select an assessment tool based on predictive performance. Our review nevertheless adds valuable information to the existing body of literature concerning the

tool that is currently most suitable for use by primary care providers to identify patients who are at high risk of falls.”.

VERSION 3 – REVIEW

REVIEWER	Ambrens, Meghan CQUniversity, School of Health, Medical and Applied Sciences
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	Well done in addressing each of the comments through this review process.